A Minimal Epistemic Core

Differentiation, Receptive Dependence, and Registration

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ABSTRACT. This paper states a minimal core of conditions that hold wherever something is taken up: perceived, or received in any way, by anything, whatever the domain. The core is not itself an epistemic relation. It is a structure that every taking-up contains, and it is found in much that nobody would call epistemic; what makes a taking-up epistemic is no part of it. The account starts from two commitments and one decision. The first commitment is that what is taken up is something in particular, and is taken up as it differs from something else. The second is that the taking-up holds, at least partly, in virtue of what is taken up. The decision is that the account uses only differences between actual terms, never differences from alternatives; its reasons and its cost are stated. On this basis registration is defined: one contrast between actual terms holds at least partly in virtue of another. The definition presupposes neither understanding, nor representation, nor order, nor a pairing of terms, nor a knowing subject, nor any repertoire of possibilities. The account uses two notions without analysing them, a field's bearing a contrast in a respect and one contrast's holding in virtue of another, and it states the conditions that bind each. The conditions are formal: they presuppose no spatial, temporal, or physical structure. They are claimed to be necessary for any taking-up, not sufficient, and the claim is no stronger than the two commitments. Two cases, one causal and one not, show that the structure is met by dependences of different kinds. The paper lists everything it assumes, by kind, and what the core does not provide.

1. Introduction

Theories of knowledge ordinarily begin with a subject who believes, judges, or is justified, and ask under what conditions such states amount to knowledge. This paper addresses a prior question: what must hold for anything to be taken up at all, that is, perceived, or received in any way, by anything. Its answer is a set of necessary conditions, not a sufficient account. It identifies what any theory of perception presupposes, and what theories of belief and knowledge presuppose so far as belief and knowledge involve something's being taken up;

it leaves to such theories what makes a relation epistemic. The assessment of judgement, in particular, lies outside its scope.

The phrase *taken up* is used for the subject of the inquiry before any theory of it, and it is not analysed. The word *registration* is reserved for the structure defined in Section 7. The claim of the paper is that every taking-up contains a registration. The converse is false: the structure is found where nothing is taken up by anything (Section 8). The core is therefore not an epistemic relation of a simple kind. It is what every taking-up contains.

Not every epistemic relation is covered. A claim may concern something by description, a future event, or a mathematical object. Whether such relations involve something's being taken up is left open (Section 10).

The conditions are formal. They are stated without reference to space, time, matter, or causation, and they are meant to hold whatever the domain in which something is taken up. Examples drawn from particular domains show how the conditions are met there; they do not fix what the conditions are.

The account is a reconstruction, in this sense: each element is introduced only at the point where the account cannot proceed without it, and each is accompanied by a statement of what it does not establish. Every item is given its kind: commitment, decision, notion used without analysis, condition of method, stipulation, assumption, consequence, or presupposition. One caution applies throughout. The exposition necessarily employs ordinary logical resources—identity, distinction, relation, and inference—but using them to state the account does not attribute them to what the account describes, and in particular not to a knowing subject. A distinction drawn in the exposition is not thereby a distinction drawn by its object. In the same way, the exposition uses modal words to say what the conditions leave open; this attributes no possibilities to what it describes.

The proposal has close neighbours. Dretske (1981) analyses the information carried by a signal in terms of nomic dependence and conditional probability, and separates it from meaning and belief. Floridi's diaphoric definition of data takes a lack of uniformity—a difference—as the most elementary constituent of information (Floridi 2011), and his informational structural realism holds that the relation of difference is constitutive of its relata (Floridi 2008). Smith's metaphysics of presence makes *registration* its central notion, as the way in which a world comes to be taken as one of objects (Smith 1996).[1] Enactivist accounts describe cognition as the structural coupling of a system with a medium, both of which belong to a larger organization (Maturana and Varela 1987; Thompson 2007). The present account differs from each in three respects. It separates the contrast that is registered, the contrast that registers it, and the dependence that connects them. It states that dependence between contrasts of actual terms,

1. The present use of the term is narrower. Smith's registration is performed by subjects and yields objects; registration here presupposes neither and concerns only the dependence of one contrast on another.

without probabilities, causes, or alternatives on either side, so that it applies beyond the physical domain and requires no repertoire of possibilities. And it distinguishes the direction of dependence from any order of occurrences, leaving open what kind of order further epistemic operations require.

Section 2 states the two commitments and the decision, in ordinary words. Sections 3 to 6 introduce the vocabulary in which they are made exact: fields and contrasts, articulations, the dependence of one contrast on another, and the receiving field. Section 7 defines registration and shows how every taking-up contains one. Section 8 gives two cases, and Section 9 considers order, which the structure does not assume. Section 10 lists everything the account assumes, what it rules out and what it does not include, what it leaves undone, and the questions it leaves open.

2. Two Commitments and a Decision

The account starts from two commitments about what it is for something to be taken up, and from one decision about what the account itself will use.

> (D) DIFFERENTIATION
>
> *What is taken up is something in particular, and it is taken up as it differs, in some respect, from something else.*

The commitment has two parts. The first is that a taking-up is of something in particular: there is a fact about what is taken up. This is assumed. The second is that the thing is taken up as it differs. The reason for it lies in what "in particular" requires. For a thing to be some way in particular, there must be a way that it is not, and its being this way as against another lies in that difference. What is not one way as against any other is no way in particular, and cannot be what a taking-up is of. A taking-up of something as the particular thing it is, is then a taking-up of it as it differs. This is a reason, not a proof: it supposes that to take up something in particular is to take it up in what makes it particular.

Two things are not said. It is not said that a thing is nothing but its differences. Whether a thing's character consists in how it differs, or is more than that, is left open; the commitment concerns the particularity of its way, not what the thing is. Nor is it said, in (D), that what a thing differs from must be actual. That is the matter of the decision below.

(D) can be refused in two ways: by holding that a taking-up need not be of anything in particular, or by holding that something can be taken up as the

particular thing it is without its difference from anything being taken up.

(C) RECEPTION

The taking-up holds, at least partly, in virtue of what is taken up.

This commitment says which relation the account is about: the one in which something receives. It is not a hypothesis that some case of taking-up might overturn, since a relation in which nothing held in virtue of what it is said to be of would not be a taking-up. Its content can be seen from the relations that it sets aside, which are examined in Section 5: two things may be alike, or correspond term for term, or agree because both hold in virtue of a third, without either holding in virtue of the other. "In virtue of" is one of the two notions that the account does not analyse; the conditions that bind it are given in Section 5. Since, by (D), what is taken up is something as it differs, (C) says that the taking-up holds at least partly in virtue of a difference.

(C) can be refused by holding that something can be taken up although nothing holds in virtue of it. The account regards that as a change of subject.

DECISION: ACTUAL TERMS ONLY

The account uses only differences between terms that are both actual. It uses no difference between something actual and an alternative, that is, a way that nothing actual is.

This is a decision about what the account will use, not a result. "Actual" is used without analysis, for what is the case as opposed to an alternative that is not. It is not tensed: nothing is said about when. The decision has three reasons, one for each place where alternatives might be brought in.

The first place is in what is taken up. (D) and (C) do not by themselves exclude that a thing is taken up as it differs from a way that nothing actual is. But such a way has to be supplied from somewhere. It may be supplied by a stock of possibilities, which is an assumption about what there is that the account does not make. It may be supplied by what receives, which is the third place below. Or it may be supplied by the account's own specification of it, and then the difference is one that holds partly because the account has specified it, which the condition of method excludes (Section 3). With an actual other term nothing has to be supplied: both terms are actual. This reason can be refused by admitting a stock of possibilities.

The second place is in the dependence. One might say that a taking-up holds in virtue of what is taken up when it would not have held had that been otherwise. But the two come apart. One thing may hold in virtue of another although, had the second not held, a third would have brought the first about. And it can be that neither of two things holds without the other although neither

holds in virtue of the other, because both hold in virtue of a third. What would have held otherwise is therefore evidence of dependence, often the best there is, and in this simple form it is not what dependence consists in. More refined analyses that also use alternatives are not examined here; the account does not state the dependence over alternatives.

The third place is in what receives. Alternatives might be held there as a repertoire, against which the actual case is singled out. The structure defined in Section 7 uses no repertoire. A repertoire would also bring in a relation of another kind: its members would have to stand for what is not the case, and to stand for what is not the case is not to hold in virtue of it. Whether some kinds of taking-up require a repertoire is not settled here. If they do, it enters as a further commitment, stated as such.

The decision applies to both sides. What is taken up is treated as a difference between actual terms, and so is that by which it is taken up (Section 6).

The decision has a cost. The core does not provide for taking up which way something is as against an alternative; anything of a term that differs from nothing actual in the respect concerned; or that two terms do not differ; and a receiver with one condition only is not covered (Section 6). Nothing is claimed about whether these occur. If they do, they need what the core does not contain.

What kind of claim this is. The two commitments are an analysis of what it is for something to be taken up as the particular thing it is. If the analysis is right, the conditions that follow hold of every taking-up, of whatever kind, and nothing in them concerns what sort of thing receives. The necessity claimed is of that kind and no stronger: it is the necessity of the analysis, and it holds given the decision. It is not claimed that anything is ever taken up, nor that the conditions hold of anything apart from a taking-up. It may be objected that reasoning of this form shows only how we must think of epistemic relations, not how they are (Stroud 1968). The commitments do not pass from how we think to how things are. They say what a relation must be in order to be a taking-up of something, and they leave open whether, where, and by what anything is taken up. What remains true in the objection is that an analysis is answerable to the notion analysed, and that we know that notion from our own case. That is why the ways of refusing each commitment are stated.

"In particular" does not mean "one only". The core does not single out one difference (Section 5).

3. Fields and Contrasts

The first notion that the account uses without analysing it is that something *bears a contrast* in some respect: two of its terms differ in that respect. It is left unanalysed because any definition of it would use the notion of a difference, or one no clearer. A notion used in this way must still be bound by stated

conditions, so that a misuse of it can be shown. There are three. A contrast is always in some respect. It is between two terms, and no term differs from itself. And, by the decision of Section 2, both terms are actual.

Two remarks are added, which are not conditions. The account does not say what respects there are, nor whether bare distinctness is one; if it is, any two things bear a contrast, and (D) asks correspondingly little. And a contrast does not divide what bears it exhaustively: its two terms need not be all there is to what bears it.

Whatever bears a contrast is, insofar as it does, a *field*. To be a field is thus an aspect, not a kind of thing: the term implies neither a spatial region, nor a collection of discrete objects, nor something already presented to a subject. What it is for terms to be the terms of one field is not analysed, and the account does not say how fields are individuated. Let F designate the field under consideration. The designation fixes the scope of the inquiry. It does not establish a boundary intrinsic to reality: F identifies what the account concerns, and any claim about its actual boundaries would require separate justification.

By stipulation, two contrasts are the same when they have the same terms and the same respect. The same two terms may therefore bear two contrasts, in two respects.

That F bears a contrast is not a modal claim. It does not say that F can be differentiated, or that it admits differentiations which it does not have; it says that there is a difference. Nothing is thereby said about which contrasts F bears, about whether some differentiation of F is uniquely privileged, or about whether what differs would be objects, qualities, regions, values, or something else. Nor is anything implied about change or about any order. Possibilities and capacities of a field are not thereby denied. They are not part of the core, and if a later stage of the theory needs them, they must enter as commitments of their own, stated as such.

The account is subject to one condition of method.

SCOPE OF THE ACCOUNT

The account concerns only contrasts that a field bears, and none that holds, even partly, because the account has specified it.

The reason is that the account concerns a differentiation attributable to its target, rather than one supplied entirely by an unrelated description; a contrast is attributable to a field when the field bears it. This is a necessary condition for the route of explanation pursued here, and it agrees with (C): a difference that held only because of a description could be taken up only as a difference in the description. The account does not argue against the view that all differentiation is supplied by descriptions. It states what that view would cost: if it held, no contrast would be attributable to a target, and the account would have nothing

to articulate. No knowing subject is involved: that F bears a contrast does not imply that anything differentiates it.

4. Articulation

An account speaks of a particular contrast by means of an *articulation*, written

$$\langle a, \delta, b \rangle_F,$$

which names two terms, a and b, and a *respect*, δ. An articulation is *specified* when its terms and respect are fixed, and it is the account, not F, that specifies it. It is *realized* when the terms it names are actual terms of F and differ in the respect δ. The *contrast* of a realized articulation is the difference of a and b in that respect; two articulations may share a respect and have different contrasts. An articulation can fail to be realized in two ways: one of its names may pick out nothing, or the terms it names may not differ in the respect named.

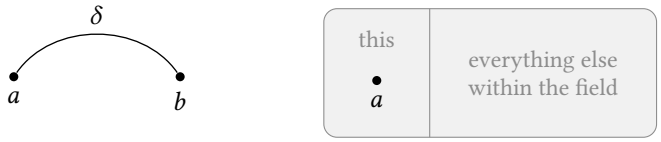
The terms of a realized articulation are both actual. They are not alternatives of which one is the case and the other not, and the contrast is not a choice between them; it is the difference between two terms that are both actual. The schema has two terms because a contrast is between two terms; a differentiation among more terms is a number of contrasts. The schema does not assume that a and b exist prior to the contrast as independent objects, and it does not deny it.[2]

Two misreadings of the schema should be excluded. First, δ is not a third item situated between a and b. To treat it as a separator would presuppose spatial organization, whereas δ is the respect in which the two differ. Second, b need not comprise everything that remains once a has been selected. That reading would require an exhaustive partition of F, and a contrast between two terms is not equivalent to a division into "this" and "everything else within the field": the latter adds the requirement that every case within the scope fall on one side or the other. Figure 1 sets the two constructions side by side.

Specification chooses the respect and selects a contrast. By the condition of Section 3, it does not create the contrast. Whether F realizes the specified articulation, and whether anything holds in virtue of its contrast, are not settled by the specification, which can therefore fail. Specification fixes what an inquiry concerns, as F fixes its scope; it does not supply what the inquiry finds.

The same condition leaves out any contrast that holds, even partly, because of the specification or its notation: because of the order, the labels, or the

2. That a difference need not be grounded in terms identified before it is argued by Deleuze in his critique of the subordination of difference to identity (Deleuze 1994, ch. 1); the present account leaves the question open. Deleuze also rejects the possible as a mere image of the real, and the present account likewise uses no stock of possibilities.



(a) contrast (b) exhaustive partition

FIGURE 1. Two constructions over F. (a) An articulation $\langle a, \delta, b\rangle_F$: a and b mark terms, both actual, and δ, drawn as a relation rather than as an item between them, is the respect in which they differ. (b) An exhaustive partition, which adds the requirement that every case fall on one side or the other. Only (b) is drawn as a region, because only (b) carries the extensional commitment that a region suggests; in (a) the placement of the marks has no spatial meaning.

grouping of the terms. Suppose that a and b differ in the respect δ, and let δ^* be the respect: δ, together with being written first. That a and b differ in δ^* holds partly because of how the account has written them. It is left out for that reason, and not because something other than F has a part in it: a contrast of F may well hold partly in virtue of a contrast borne elsewhere (Section 5). A differentiation of this mixed kind is not left out wholesale. The contrast in δ is within the scope of the account; the remainder is not.

Two things follow from the definitions. First, a field in which an articulation is realized bears that articulation's contrast. This establishes neither which contrasts a field bears, nor that any field bears a contrast apart from a taking-up, nor any modal claim about F. Second, what differs from no actual other term in a given respect is a term of no contrast in that respect, and the core provides for taking up nothing of it in that respect.

That a field bears contrasts does not fix which articulation an account will specify. An account of any particular case must therefore specify the articulation it concerns, and it may fail to pick out anything. And a realized articulation is an articulation of F: nothing so far implies that its contrast is taken up by anything.

5. Receptive Dependence

That F bears a contrast, and that some other contrast is borne somewhere, leaves the two unconnected. Commitment (C) requires a connection: that the one hold in virtue of the other. "In virtue of" is the second notion that the account does not analyse, and this section gives the conditions that bind it. The account also speaks of one contrast *depending* on another, and of *receptive dependence*; these are the same notion. Where it says, more loosely, that something holds because of something that is not a contrast, the word is the ordinary one, and none of the conditions is claimed for it.

The notion is left unanalysed for the following reason. The analyses available either state the dependence over alternatives to the actual case, as counterfactual analyses do, which the account has decided not to do (Section 2); or belong to one domain, as causal analyses do; or identify it with metaphysical ground. The account does none of these. From the literature on ground it takes two features only: that dependence of this kind has a direction, and that it relates what actually holds (Fine 2012).[3]

Three conditions bind the notion. First, it relates contrasts that both hold. A statement of dependence is therefore not a statement about what would hold otherwise. Second, nothing holds in virtue of itself. This is stipulated, so that no contrast depends on itself. Third, the dependence has a direction: that K' holds in virtue of K is one thing, that K holds in virtue of K' is another, and the first does not bring the second with it. The conditions are stated for the case in which both things related are contrasts, which is the case that the definition of registration uses. (C), as stated in Section 2, relates a taking-up to what is taken up; Section 6 says how the account brings that under a relation between two contrasts.

Three further matters are permitted or left open; they are not conditions. The dependence may be partial: K' may hold in virtue of K and of other things as well, which may lie in the field that bears K'.[4] Whether two distinct contrasts can each hold partly in virtue of the other is left open. (C) does not exclude it: it requires that the taking-up depend on the difference, not that the difference be independent of the taking-up. A view on which a difference in F is partly made in being taken up is described, in these terms, as two contrasts each depending on the other. Whether the dependence is transitive is also left open. The two questions are connected. If the dependence is transitive, two contrasts that depended on each other would each depend on itself, which the second condition forbids. So transitivity excludes mutual dependence, and mutual dependence excludes transitivity. The account does not decide between them. If the dependence is transitive, it is a strict partial order among contrasts; if it is not, loops through a third contrast are not excluded.

Dependence is to be distinguished from three relations that can hold where neither contrast holds in virtue of the other. Two contrasts may resemble one another.[5] The terms of one may be paired with the terms of the other, each

3. Counterfactual analyses of dependence require alternatives to the actual case. Where contrasts hold necessarily, they meet a further difficulty: on the standard semantics, counterfactuals with impossible antecedents are all vacuously true (Lewis 1973); for an alternative semantics, see Nolan (1997).

4. This accommodates the enactivist point that a perturbation triggers, but does not specify, the changes it occasions in a system, which are fixed by the system's own structure (Maturana and Varela 1987).

5. Goodman (1976, ch. 1) makes the parallel point about representation: resemblance is reflexive and symmetric, whereas representation is neither, so resemblance cannot

with each (Figure 2). And two contrasts may agree because both hold in virtue of a third. In the first two cases the relation can be read in either direction; resemblance holds equally between a contrast and every other that is like it, and a pairing can be set up between any two contrasts. In the third, whatever connection there is lies in the dependence of each on the third. None of these is what (C) requires, and (C) requires none of them. In particular it requires no pairing of terms: it relates one contrast to another, and which term of the one answers to which term of the other is not part of it. Where a pairing is needed, it must enter as a further commitment, stated as such.

The dependence so bound is close to what Bateson's definition of information as "a difference which makes a difference" expresses (Bateson 1972); MacKay had earlier described information as a distinction that makes a difference (MacKay 1969). The phrase compresses what the present account keeps apart: the first difference is K, the second is K', and the making is the dependence. The dependence should not be replaced by talk of "systematic" or "relevant" coupling. Such terms bring in a standard of relevance, or a regularity across cases, which the account has not supplied.

Whether K' holds in virtue of K need not be fixed by a description of the two contrasts. In some kinds of case it is fixed; the second case of Section 8 is one. In others it is not: two pairs of contrasts alike in every description, and in any pairing of their terms, may differ in that in one pair the second contrast holds in virtue of the first and in the other pair it does not. Whether, in cases of that kind, the dependence in one instance holds because of other actual instances, as in a regularity, is left open.

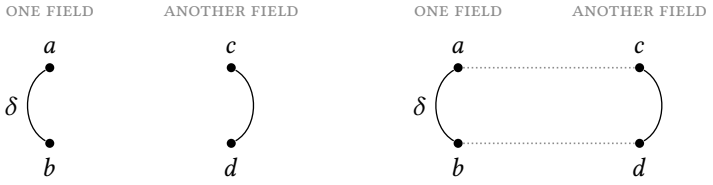
The dependence does not single out one contrast. If K' holds partly in virtue of K and partly in virtue of another contrast K^*, it depends on both; and if the dependence is transitive, it depends on whatever K depends on. In the sense of registration defined in Section 7, a contrast registers every contrast it depends on. Which of these a taking-up is of is not settled by the core.

The direction of the dependence defines two roles. Where K' holds in virtue of K, the field that bears K is the *target* in that instance, and the field that bears K' is the *receiver*. A field has a role only through the contrast it bears in that instance: one field may bear both contrasts, and is then target and receiver at once. Roles are thus places in a relation, not natures of what occupies them, and the same field may be a target in one instance and a receiver in another.

Two limits of what has been said should be noted. The dependence is not derived from the existence of contrasts: a relation cannot be obtained merely by describing what it relates. It is what (C) commits the account to. And the dependence is not analysed. What it consists in within a particular domain is a question for that domain: in physical domains it may be causal, in others logical

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be what representation consists in.



(a) resemblance (b) pairing

FIGURE 2. Two relations in which neither contrast holds in virtue of the other. (a) Resemblance: two contrasts are alike. (b) Pairing: a is paired with c, and b with d. The pairing is drawn without direction, since it can be read either way. Neither panel shows a dependence, and a dependence requires neither.

or constitutive. The core is in this sense a schema. It states the conditions that a dependence satisfies, and each domain supplies the dependence.

6. The Receiving Field

(C) speaks of the taking-up as holding in virtue of what is taken up. This section says what the account takes the taking-up to be.[6]

The taking-up is some way that what receives is. The account assumes that it is some way in particular. By the reasoning given for (D), its being that way in particular lies in its differing from another way; and by the decision of Section 2, the account treats that other way as actual too. The account therefore takes the taking-up to be one term of a contrast between two actual *conditions* of what receives, written r and r', and it calls that contrast the receiving contrast, K'. The other condition, r', need not hold in virtue of anything in the target. It is only that from which r differs.

This is the least secure step of the account, and it is marked as such. Nothing in the two commitments requires that what receives have two actual conditions. That it does is the decision for actual terms, applied to the receiving side as to the target; a receiver with one condition only, differing from nothing actual, is not covered. And one further assumption is used, which is not argued: that a contrast holds at least partly in virtue of whatever one of its terms is what it is in virtue of. With it, K' holds at least partly in virtue of what is taken up.

(R) THE RECEIVING CONTRAST AND ITS BEARER

Where a contrast K' holds in virtue of a contrast K, K' is distinct from

6. The vocabulary of reception recalls Kant's characterization of sensibility as receptivity, the capacity to acquire representations through being affected by objects (Kant 1998, A19/B33, cf. A50/B74). The receiver introduced here is thinner: it involves no representations, and nothing is claimed about forms of intuition.

> *K. What bears K' is, insofar as it does, a field; R designates it, and it may be the field that bears K.*

(R) is not a further commitment. That K' is distinct from K follows from the stipulation that nothing holds in virtue of itself, and that its bearer is a field follows from what a field is. F and R are letters for the two roles in the instance considered; nothing else is meant by them.

R is a field in exactly the sense in which F is one: whatever bears a contrast, insofar as it does. Bearing a contrast is not a property peculiar to receivers. What holds of the target therefore holds of R: the contrast concerned is attributable to the field that bears it, and it is the account that picks it out. R itself distinguishes nothing. In particular, it does not distinguish a taking-up from its absence: that K' holds in virtue of K is a fact about the two contrasts, not something that R notes.

R is not a person, a conscious subject, or an independently bounded system. Nothing follows about whether R and F are distinct fields, parts of a larger one, or a single field bearing two contrasts, since a distinction of contrasts does not entail a distinction of substances. A field may be target and receiver at once, but only through two contrasts.

As with the terms of a realized articulation, r and r' are both actual: they are not alternatives between which R is poised. Nothing requires that K' be like K in any respect, and no condition is assigned to any particular term of K.

Neither contrast is given with the other. Nothing in what a contrast of F is implies that some contrast holds in virtue of it, and a field may bear contrasts that hold in virtue of nothing in F. A field's merely differing within itself does not make it a receiver of anything in F.

This fixes how much individuation the account admits. There must be enough to distinguish K from K', and to take r and r' as terms of one contrast. There need not be enough to identify enduring objects or successive states of a continuing subject. To treat these stronger forms of individuation as already established would build into the foundation a persisting, bounded bearer—which is precisely what an account of a knowing subject must eventually explain.

7. Registration

Registration can now be defined, in a restricted, structural sense.

DEFINITION OF REGISTRATION

> *A contrast K is registered by a contrast K' when K' holds at least partly in virtue of K. K is then the* registered *contrast and K' the* registering *contrast, and the field that bears K' is said to register K.*

Registration is always of a particular contrast: to say that R registers F without qualification is to leave the claim incomplete. An account that states a registration must therefore specify an articulation; the specification is the account's, and it is no part of the registration. Registration is an actual relation between contrasts that hold, not a standing disposition of R to respond. Figure 3 depicts the structure so defined.

The claim of the paper can now be stated exactly: every taking-up contains a registration. By (D), what is taken up is something as it differs from something else, and by the decision that is a contrast K between actual terms. By (C), the taking-up holds at least partly in virtue of K. By Section 6, the account takes the taking-up to be one term of a contrast K' between actual conditions of what receives, and K' then holds at least partly in virtue of K. So K is registered by K', and the receiving contrast of Section 6 is the registering contrast. The three steps differ in kind. The first uses a commitment and the decision. The second uses a commitment. The third uses the decision, applied to the receiving side, and two assumptions that are not argued: that the taking-up is some way in particular, and that a contrast holds partly in virtue of whatever one of its terms is what it is in virtue of.

The converse does not hold. A registration is not, as such, a taking-up of anything, and Section 8 gives a case. The core says nothing about what must be added to a registration for something to be taken up.

The definition attributes to the receiver no understanding, identification, or judgement, since none of these capacities has been introduced. Where registration is realized physically, it is found in systems to which nobody would attribute knowledge, and by itself it does not place anything "in the logical space of reasons".[7] The definition does not require that anything pass from F to R, nor that the registering contrast be like the registered one in any respect. Nor does it require a representational space, or any repertoire of alternatives, in R. Registration relates contrasts that hold, and nothing in R need stand for conditions that R does not have.

Registration so defined is supplied by neither field in isolation. It comprises three distinct contributions: from the target, the registered contrast; from the receiver, the contrast between its conditions; and between them, the dependence that connects the two. Keeping these contributions distinct prevents registration from being attributed to the target merely because the target is differentiated, or to the receiver merely because its conditions differ. This does not explain the emergence of a knowing subject, but it identifies what such an explanation would have to elaborate.

7. The phrase is from Sellars (1956, § 36); Brandom (1994, ch. 4) develops the contrast between such reliable differential responsiveness and conceptual capacities. In perception, Dretske's non-epistemic seeing, in which an object is visually differentiated from its surroundings without any belief about it, is a case that contains a registration in the present sense (Dretske 1969).

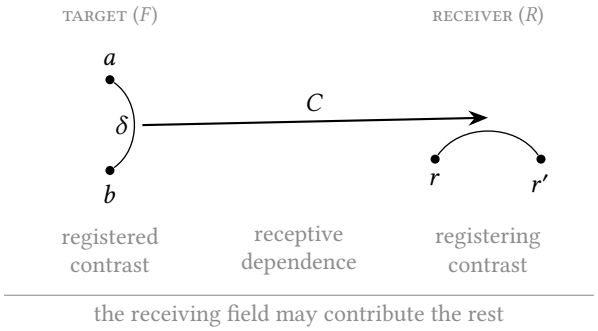


FIGURE 3. Registration. The contrast between the conditions r and r' of R holds at least partly in virtue of the contrast of the realized articulation $\langle a, \delta, b\rangle_F$. All four terms are actual. The registering contrast is drawn with a different organization, since it need not be like the registered one in any respect, and no term is paired with a condition. The arrow runs from contrast to contrast and denotes receptive dependence only: it represents neither a passing of anything from one side to the other nor an order of occurrences, and the columns are roles, not regions.

What is registered is always a difference between actual terms. The core therefore does not provide for registering which way a term is, as against a way that nothing actual is; anything of a term that differs from nothing actual in the respect concerned; or that two terms do not differ in a respect, since where there is no contrast there is nothing to register. Accounts of information and knowledge that start from alternatives presuppose a set among which the actual case is singled out.[8] The core uses no such set, by the decision of Section 2. How the three things just listed come about, where they do, is left to a later stage. Registering which way a term is appears to need at least a pairing of terms with conditions, which registration does not include.

8. Two Cases

The core is a schema, and a schema should be shown to have instances. Two are given here, chosen so that the dependence is of a different kind in each. They test the schema; they do not fix what it says, and no argument in the preceding sections relies on them.

A causal case. Let a and b be two bodies that differ in mass, resting on the two pans of a balance that is at rest, with its ends at different heights. K is

8. Shannon's theory measures information as selection from a set of possible messages (Shannon 1948), and Dretske's informational content is defined by conditional probabilities over such alternatives (Dretske 1981). In epistemology, relevant-alternatives theories and contrastivism analyse knowledge as the elimination of alternatives (Dretske 1970; Lewis 1996; Schaffer 2004); counterfactual and interventionist analyses of dependence share the same structure (Lewis 1973; Woodward 2003).

the contrast of a and b in mass; K' is the contrast of the two ends in height. Both hold, and all four terms are actual. K' holds at least partly in virtue of K, and the dependence is causal. It is partial: K' holds also because of how the balance is built. K does not hold in virtue of K'. The beam has no repertoire of the weights that the bodies might have had. The case shows that registration does not single out one contrast: K' holds partly in virtue of the difference in where the two bodies rest on the beam, and so registers that contrast as well. Something passes from the bodies to the beam in this case, and the dependence came about through a process in time. The core neither requires nor excludes either.

A constitutive case. Let δ be a fine respect and Δ a coarser one, such that terms which differ in δ in a certain way differ, for that reason, in Δ; a finer and a coarser classification of the same terms are an example. Let a and b be actual terms that differ in δ in that way. K is their contrast in δ, and K' is their contrast in Δ. They are two contrasts, since the respects differ, although the terms are the same. K' holds in virtue of K, and K does not hold in virtue of K'. The dependence is not causal, and nothing in it is temporal or spatial. Three things differ from the first case. K' holds in virtue of K alone, given the two respects, so the schema does not require that what receives contribute anything. Whether K' depends on K is fixed by a description of the two. And one field bears both contrasts: it registers one of its own contrasts through a second.

Three things follow. First, the schema is met by dependences of two kinds. Second, in each case the dependence is identified by the means of its own domain. Those means may include the comparison of other cases, or of what would happen otherwise; the conditions of the core say what the dependence relates when it holds, not how it is found to hold. Third, registration is not itself epistemic. In the second case nobody would say that anything is taken up, and the schema is met. Wherever one difference holds in virtue of another there is a registration, so registrations are very common. This is consistent with a statement of necessary conditions, and it fixes how the result is to be read. The core states what every taking-up contains, and much else contains the same. What must be added for something to be taken up is not said by the core.

9. Registration and Order

The core assumes no order of any kind. Its contrasts are not sequences. It does not say that the conditions of R are successive, and it does not say that they are not: "actual" is not tensed, and whether what is earlier and what is later are both actual is not settled here. The dependence has a direction, but the direction of a relation is not an ordering of occurrences: that K' holds in virtue of K does not place either before the other.

The core therefore cannot describe a receiver as *acquiring* a contrast, *retaining* it, or *comparing* it with another. What each of these appears to require is stated here for the use of later stages; it is not derived.[9] Comparison appears to require two registering contrasts that both hold, and no order. Retention appears to require that one receiver be identified across distinct places of some ordering. Acquisition appears to require that a place of such an ordering at which a contrast is not registered stand in an asymmetric relation to one at which it is. Neither retention nor acquisition follows from the definition of registration.

What they appear to require is order in the loosest sense: an asymmetric relation among places, with no requirement that any two places be comparable, and with no metric, no linearity, and no universal index. Whether this relation must also be transitive, and so form a strict partial order, may be left open. Temporal succession is one way of realizing such an order; the order of steps in an inference or a proof is another, which is not temporal. The core is compatible with either and requires neither. If the dependence of Section 5 is transitive, it is itself an order of this kind among contrasts; whether it can serve here is not examined.

Since the direction of a relation is not an ordering of occurrences, if receptive dependence is written with a directed symbol, such as an arrow from F to R, that symbol should not also serve, without further explanation, to represent order. The direction of dependence and the asymmetry of an ordering play different roles in the account and should be kept notationally distinct.

10. Conclusion

Table 1 lists what the account assumes, by kind, with what each item leaves unestablished.

The result is that every taking-up, in whatever domain, contains a registration: one contrast between actual terms holding at least partly in virtue of another, with target and receiver distinguished as roles within that dependence. The conditions are necessary for a taking-up and not sufficient. Their necessity is that of the analysis given in the two commitments, and it holds given the decision. Nothing is claimed about relations to things by description.

The core is minimal, but it is not empty. It rules out one thing: that a taking-up consists in resemblance, in a pairing of terms, or in an agreement that two

9. The three operations closely parallel the synthesis of apprehension, reproduction, and recognition in the first edition of Kant's Transcendental Deduction (Kant 1998, A98–A110), and Husserl's distinction between retention and recollection (Husserl 1991) analyses the second of them. Both treat the operations as presupposing temporal form; the present account speaks only of order, of which temporal order is one kind.

TABLE 1. What the account assumes, by kind, and what each item leaves unestablished.

ITEM	KIND	WHAT IT LEAVES UNESTABLISHED
(D) What is taken up is something in particular, taken up as it differs from something else	Commitment; first part assumed, second part with a stated reason	Whether a thing is more than its differences; whether what it differs from is actual.
(C) The taking-up holds at least partly in virtue of what is taken up	Commitment; says which relation the account is about	That anything is taken up; which contrast, among those depended on, a taking-up is of.
Actual terms only, on both sides	Decision, with three reasons	Which way something is as against an alternative; anything of a term that differs from nothing actual in the respect concerned; that two terms do not differ; a receiver with one condition only. Also: that there are no possibilities.
Bearing a contrast in a respect	Used without analysis; three conditions	Which contrasts a field bears; what respects there are; how fields are individuated; any capacity of a field.
Holding in virtue of	Used without analysis; three conditions	What it consists in within a domain; whether it is transitive or can be mutual; whether it is fixed by description.
Scope: no contrast that holds because the account has specified it	Condition of method	That no difference is supplied by a description.
When two contrasts are the same; that nothing holds in virtue of itself	Stipulated	Nothing further.
That the taking-up is some way in particular; that a contrast holds partly in virtue of whatever one of its terms is what it is in virtue of	Assumed, not argued	Nothing further.
(R); the lone term; that every taking-up contains a registration	Consequences	That a registration is a taking-up; that R and F are distinct in substance.
"Taken up", "epistemic", "actual"; the logic and the modal words of the exposition; the account's own reference to its targets	Presupposed	Not examined here.

contrasts owe to a third, since in none of these does one contrast hold in virtue of the other (Section 5). It leaves out of the account, as a condition of method, any contrast that holds even partly because of the account's own specification (Sections 3 and 4). And it does not include, among the conditions of registration, any of the following: that something pass from target to receiver; a repertoire of alternatives; time, or order of any kind; a pairing of terms with conditions; a knowing subject, concepts, or representations. That these are not among the conditions does not show that a taking-up can occur without them in a given domain, and it does not deny that they are present in particular epistemic relations. What is claimed is only that the two commitments do not require them: wherever they are needed, they must enter as further commitments, stated as such.

What the account leaves undone should be stated as plainly. First, the two notions it uses without analysis are bound by stated conditions and no more. The distinction between a contrast of the target and one supplied by a description rests entirely on the first of them, and the account does not say what respects there are or how fields are individuated. Second, the core does not say what must be added to a registration for something to be taken up; registrations are very common, and most are not epistemic. Third, by the decision for actual terms the core does not provide for taking up which way something is as against an alternative, anything of a term that differs from nothing actual in the respect concerned, or that two terms do not differ; and a receiver with one condition only is not covered. Fourth, the treatment of the receiving side (Section 6) rests on that decision and on two assumptions that are not argued. Fifth, registration does not single out one contrast, and since transitivity is left open, the core does not say whether a registering contrast depends on more than what it immediately depends on. Sixth, the core says by what conditions receptive dependence is bound, not how an instance of it is known to be one, and it sets no bound on what a receiving field contributes. Seventh, "actual" is not tensed. Whether what is earlier and what is later are both actual, and so whether there can be a contrast between them, is not settled here and belongs with the question of order. Eighth, the account picks out the contrasts of its targets by description, and neither commitment covers a relation to something by description: the account uses a relation to its targets that it does not itself cover.

Five questions remain open. The first is what the dependence consists in, and what gives it its direction, in a given domain. The second is whether it can be explained in terms of more elementary relations, and whether it is transitive or can be mutual; to write it as an arrow would not answer this. The third is what must be added to a registration for something to be taken up, and how taking up which way a term is, anything of a lone term, and likeness come about. The fourth is what order acquisition and retention require, whether anything stronger than an asymmetric relation is needed, and whether the dependence

itself, if it is transitive, supplies that order. The fifth is whether relations to something by description, to the future, and to mathematical objects involve something's being taken up. The account ends at registration. Order, memory, recognition, and representation have not been presupposed in reaching it, and none of them has yet been derived from it.

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